

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – 6

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Class: TYIT	Batch: 1
Date of Assignment: 30/07/2026	Date/Time of Submission: 30/07/2026

a. Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

Code:-

1. Create wwwroot/manifest.json

```
{  
  "name": "HTML TO PWA",  
  "short_name": "HTP",  
  "start_url": "/",  
  "display": "standalone",  
  "icons": [  
    {  
      "src": "icons/icon1.png",  
      "sizes": "128x128",  
      "type": "image/png",  
      "purpose": "any"  
    },  
    {  
      "src": "icons/icon2.png",  
      "sizes": "512x512",  
      "type": "image/png",  
      "purpose": "any"  
    }  
  ]  
}
```

2. Create wwwroot/service-worker.js

```
self.addEventListener("install", e => {  
  console.log("Service Worker Installed");  
});  
self.addEventListener("fetch", e => {  
  e.responseWith(fetch(e.request));  
});
```

3. Add to _Layout.cshtml

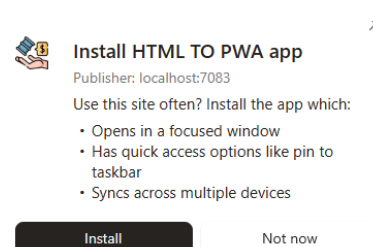
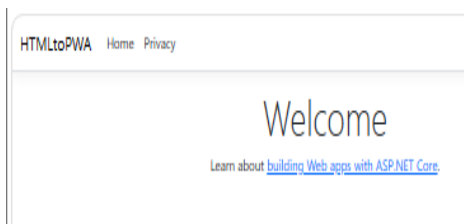
Inside the <head> section:

```
<link rel="manifest" href="/manifest.json">
```

Before the closing </body> tag:

```
<script>  
if ('serviceWorker' in navigator) {  
  navigator.serviceWorker.register('/service-  
worker.js');  
}  
</script>
```

Output:-



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b. Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:

Offline.html

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8" />
  <title></title>
</head>
<body>
  <h2> offline </h2>
</body>
</html>
```

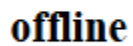
Service-worker.json

```
const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offline.html"
      ]);
    })
  );
});

console.log("Service Worker Installed");
});

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() => caches.match("/offline.html"))
  );
});
```

Output:



offline